Day 2 — Java Programming Fundamentals

Topic:-

- Introduction to Java:

Java is a **high-level, object-oriented programming language** used to build desktop applications, web applications, Android apps, and enterprise software.
Java was developed by **James Gosling and his team at Sun Microsystems** and released in 1995. Today, Java is maintained by Oracle and the Java community.

1. Features of Java:-

- **Simple:** Its syntax is relatively easy to learn.
- **Object-oriented:** Programs are organized using classes and objects.
- **Platform-independent:** Java code can run on different operating systems using the Java Virtual Machine (JVM).
- **Secure:** It provides features that help build secure applications.
- **Robust:** It supports exception handling and automatic memory management.
- **Multithreaded:** It can perform multiple tasks concurrently.

- JDK, JRE, and JVM:-

JDK, JRE, and JVM are three important concepts in Java. They work together to develop and run Java programs.

1. What is JVM?

JVM (Java Virtual Machine) is a virtual machine that executes Java bytecode.

- It converts bytecode into machine instructions using interpretation and/or JIT compilation.
- It manages memory.
- It performs garbage collection.
- It helps Java programs run on different operating systems.

Example: When you run a compiled Java program, the JVM executes its bytecode.

1. Features of Java

- **Simple:** Its syntax is relatively easy to learn.
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- **Platform-independent:** Java code can run on different operating systems using the Java Virtual Machine (JVM).
- **Secure:** It provides features that help build secure applications.

- **Robust:** It supports exception handling and automatic memory management.
- **Multithreaded:** It can perform multiple tasks concurrently.
- Java program:-

```
Public class Hello {
    Public static void Main(String[] args)
 {
        System.Out.Println("Hello, World!");
    }
}
```

Output:

```
Hello, World!
```

5. How to run a Java program

First, install the **JDK (Java Development Kit)**. It includes the tools needed to compile and run Java programs.
Open a terminal in the folder containing `Hello.java`, then run:

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- It helps Java programs run on different operating systems.

Example: When you run a compiled Java program, the JVM executes its bytecode.

2. What is JRE?

JRE (Java Runtime Environment) provides the environment needed to run Java applications.

It traditionally includes:

- JVM
- Java class libraries
- Other supporting runtime files

Purpose: To run Java programs, rather than develop or compile them.

3. What is JDK?

JDK (Java Development Kit) provides the tools needed to develop Java applications.

It includes:

- Java compiler (`javac`)
- Java launcher (`java`)
- Debugging and other development tools
- Runtime components and libraries

Purpose: To write, compile, debug, and run Java programs.

• Variables and data types:-

Variables and data types are the basic building blocks of Java programming. They are used to store and work with information such as numbers, text, and true/false values.

Variable in Java:- variable is a named memory location used to store data that a program can access and, when allowed**,** change.

Types of variables in Java

Java has three commonly discussed types of variables.

. Local variable

Declared inside a method, constructor, or block. It must be initialized before it is used.

```
Public class Demo {

void display() {
    int number = 10;
    System.out.println(number);
}
```

2. Instance variable

Declared inside a class but outside methods, without the `static` keyword. Each object has its own instance fields.

```
 Public Class Student {
    Int age = 20;
}
```

3. Static variable (class variable)

Declared inside a class using `static`. It belongs to the class and is shared by its instances.

```
   Public Class Student {
    Static String college = "ABC College";
}
```

Data types:-

A **data type** specifies what kind of value a variable can store and which operations can be performed on it.

Java data types are divided into two main categories:

1. **Primitive data types:-**

Eight built-in types that store values directly.

`byte`, `short`, `int`, `long`, `float`, `double`, `char`, `boolean`

2. **Reference data types:-**

Variables that hold references to objects or arrays.

`String`, arrays, classes, interfaces, and enums

- Operators:-

An **operator in Java is a symbol that performs an operation on one or more values or variables.**

In simple words: Operators are used to perform calculations, compare values, assign data, and make decisions in Java programs.

For example:

```
Int a = 10;
Int b = 20;

System.out.println(a + b);

Output:-30
```

Conditional Statements in Java

A **conditional statement in Java** is used to make decisions in a program. It executes different blocks of code depending on whether a condition is `true` or `false`.

For example, if a student gets marks greater than or equal to 35, the program displays "Pass"; otherwise, it displays "Fail".

Types of conditional statements in Java

Java commonly uses four forms of conditional statements:

1. `if` statement
2. `if-else` statement
3. `if-else-if` ladder
4. Nested `if` statement

Loops in Java

A **loop in Java** is a programming structure used to execute a block of code repeatedly as long as a specified condition is satisfied, or for a specified number of iterations.

For example, if you want to print numbers from 1 to 5, you can use a loop instead of writing five separate print statements.

Types of loops in Java

Java has four commonly used loop constructs:

1. `for` loop
2. `while` loop
3. `do-while` loop
4. Enhanced `for` loop (for-each)

1. `for` loop

A `for` loop is useful when you know how many times you want to execute a block of code.

Syntax:-

```
for (initialization; condition; update) {
    // Code to repeat
}
```

2. `while` loop

A `while` loop repeats a block of code as long as its condition is true.

It is useful when you do not know in advance how many iterations will be needed.

Syntax:-

```
while (condition) {
    // Code to repeat
}
```

3. `do-while` loop

A `do-while` loop executes the code block first and checks the condition afterward.

Therefore, it always executes the block at least once.

Syntax:-

```
do {
    // Code to repeat}
while (condition);
```

4. Enhanced `for` loop (for-each)

An enhanced `for` loop is used to access elements of an array or a collection one by one.
Syntax:-

```
For (dataType variable : array) {
    // Use the current element
}
```

Input and Output & Basic Problem-Solving in Java

Input and output are fundamental concepts in Java programming. They allow a program to accept information from the user, process it, and display the result.

Basic problem-solving means using variables, operators, conditions, and loops to solve simple programming problems.

1. Input and output in Java

A. Output in Java

Java uses `System.out.println()` and `System.out.print()` to display information on the screen.

Example:

```
Public class Main {
    Public static void Main(String[] args) {
        System.out.println("Hello, Java!");
        System.out.print("Welcome ");
        System.out.print("to programming.");
    }
}
Output:-
Hello, Java!
Welcome to programming.
```

Task

- Install and configure the Java development environment.

- Write programs using variables, operators, conditions, and loops.

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int a, b, sum;

        System.out.print("Enter two numbers: ");
        a = sc.nextInt();
        b = sc.nextInt();

        sum = a + b;

        System.out.println("Sum = " + sum);
        sc.close();
    }
}
```

Output:-

```
Enter two numbers: 10 20
Sum = 30
```

1.Check whether a number is even or odd.

```
Public class EvenOddNumber

{

Int number = 15;

If((number %2)==0){

System.out.println(Number +" is even number.");

}

Else

{

System.out.println(Number +" is Odd number.");

}

}

}
```

Output:-15 is odd number

2.Find the largest of three numbers.

```
import java.util.Scanner;

public class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter three numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest number is: " + a);
        } else if (b >= a && b >= c) {
            System.out.println("Largest number is: " + b);
        } else {
            System.out.println("Largest number is: " + c);
        }

        sc.close();
    }
}
```

Example output

```
Enter three numbers: 10 25 15
Largest number is: 25
```

3.Calculate factorial.

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();
long fact = 1;

        if (n < 0) {
            System.out.println("Factorial is not defined for negative
numbers.");
        } else {
            for (int i = 1; i <= n; i++) {
                fact = fact * i;
            }

            System.out.println("Factorial of " + n + " is: " + fact);
```

```
        }

        sc.close();
    }
}
```

Output:-

```
Enter a number: 5
Factorial of 5 is: 120
```

3.Check whether a number is prime.

```
import java.util.Scanner;

public class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter three numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest number is: " + a);
        } else if (b >= a && b >= c) {
            System.out.println("Largest number is: " + b);
        } else {
            System.out.println("Largest number is: " + c);
        }

        sc.close();
    }
}
```

Example output

```
Enter three numbers: 10 25 15
Largest number is: 25
```

Calculate factorial in java

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        long fact = 1;
```

```
        if (n < 0) {
            System.out.println("Factorial is not defined for negative
numbers.");
        } else {
            for (int i = 1; i <= n; i++) {
                fact = fact * i;
            }

            System.out.println("Factorial of " + n + " is: " + fact);
        }

        sc.close();
    }
}
```

Example Output

```
Enter a number: 5
Factorial of 5 is: 120
```

Check whether a number is prime

```
import java.util.Scanner;

public class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        boolean isPrime = true;

        if (n <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i * i <= n; i++) {
                if (n % i == 0) {
                    isPrime = false;
                    break;
}
                }
            }

        if (isPrime) {
            System.out.println(n + " is a prime number.");
        } else {
```

```
            System.out.println(n + " is not a prime number.");
        }

        sc.close();
    }
}
```

Output

Find the largest of three numbers in java.

```
import java.util.Scanner;

public class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter three numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest number is: " + a);
        } else if (b >= a && b >= c) {
            System.out.println("Largest number is: " + b);
        } else {
            System.out.println("Largest number is: " + c);
        }

        sc.close();
    }
}
```

Example output

```
Enter three numbers: 10 25 15
Largest number is: 25
```

Calculate factorial in java.

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        long fact = 1;

        if (n < 0) {
            System.out.println("Factorial is not defined for negative
numbers.");
```

```
        } else {
            for (int i = 1; i <= n; i++) {
                fact = fact * i;
            }

            System.out.println("Factorial of " + n + " is: " + fact);
        }

        sc.close();
    }
}
```

Example Output

```
Enter a number: 5
Factorial of 5 is: 120
```

4.Check whether a number is prime:

```
import java.util.Scanner;

public class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        boolean isPrime = true;

        if (n <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i * i <= n; i++) {
                if (n % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            System.out.println(n + " is a prime number.");
        } else {
            System.out.println(n + " is not a prime number.");
        }

        sc.close();
    }
}
Output:-
Enter a number: 7
7 is a prime number
```

1. Reverse a number.

```
import java.util.Scanner;

public class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter three numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest number is: " + a);
        } else if (b >= a && b >= c) {
            System.out.println("Largest number is: " + b);
        } else {
            System.out.println("Largest number is: " + c);
        }

        sc.close();
    }
}
```

Example output

```
Enter three numbers: 10 25 15
Largest number is: 25
```

Calculate factorial in java

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        long fact = 1;

        if (n < 0) {
            System.out.println("Factorial is not defined for negative
numbers.");
        } else {
            for (int i = 1; i <= n; i++) {
                fact = fact * i;
            }
```

```
            System.out.println("Factorial of " + n + " is: " + fact);
        }

        sc.close();
    }
}
```

Example Output

```
text

Enter a number: 5
Factorial of 5 is: 120
```

Explanation
The factorial of a number n is the product of all positive integers from 1 to n.

For example:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

- `fact = 1` initializes the factorial variable.
- The `for` loop runs from `1` to `n`.
- `fact = fact * i` multiplies the current factorial by each number.
- The factorial of `0` is `1`.
  Note: A Java `long` can store factorials only up to `20!` without overflow.

2. *You said:*

Check whether a number is prime.

3. *ChatGPT said:*

Here is a simple Java program to **check whether a number is prime or not** using a `for` loop.

Java Program

```
java

import java.util.Scanner;

public class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        boolean isPrime = true;

        if (n <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i * i <= n; i++) {
                if (n % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            System.out.println(n + " is a prime number.");
        } else {
            System.out.println(n + " is not a prime number.");
        }

        sc.close();
    }
}
```

Example output

```
Enter a number: 7
7 is a prime number.
```

5.Reverse a number in java.

```
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int reverse = 0;

        while (n != 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }

        System.out.println("Reversed number: " + reverse);

        sc.close();
    }
```

```
    }
```

Output:-

```
Enter a number: 1234
Reversed number: 4321
```

6.Print multiplication tables.

```
public class MultiplicationTable {
    public static void main(String[] args) {
        int n = 5;

        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }
    }
}
```

Output:-

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```